

Project Initialization and Planning Phase

Date	15 March 2024
Team ID	xxxxxxx
Project Title	xxxxxxx
Maximum Marks	3 Marks

Project Proposal (Proposed Solution) template

This project proposal outlines a solution to address a specific problem. With a clear objective, defined scope, and a concise problem statement, the proposed solution details the approach, key features, and resource requirements, including hardware, software, and personnel.

Project Overview	
Objective	We aim to predict plant growth stages using environmental and farm management data. This will help farmers and agricultural experts make smarter, timely decisions for better crop yield and quality.
Scope	To visualise how factors like temperature, rainfall, soil, and farm activities affect plant growth. It will support selected crops and regions based on available data and offer stage-wise predictions to guide field operations.
Problem Statement	
Description	Farmers often follow a general schedule for crop management, which may not match real-time plant growth. This leads to missed opportunities and inefficiencies.
Impact	Mistimed actions can lower crop output, waste resources, and increase risks. A data-driven approach can solve this and boost productivity.
Proposed Solution	
Approach	We will gather and analyze environmental and management data, use prediction models to estimate growth stages, visualize everything in Power BI and use filters and visuals (like charts, bar graphs, heat maps and slicers) for deeper insights. The dashboard will be interactive and user-friendly

Key Features	Visual timeline of plant growth stages Charts showing impact of weather and management actions Predictive insights for better planning Region, crop, and time filters
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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	e.g., 2 x NVIDIA V100 GPUs
Memory	RAM specifications	e.g., 8 GB
Storage	Disk space for data, models, and logs	e.g., 1 TB SSD
Software		
Frameworks	Python frameworks	e.g., Flask
Libraries	Additional libraries	e.g., scikit-learn, pandas, numpy
Development Environment	IDE, version control	e.g., Jupyter Notebook, Git
Data		
Data	Source, size, format	e.g., Kaggle dataset, 10,000 images